VANSHIKA GURBANI

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Engineer building intelligent systems at the intersection of LLMs, full-stack software, and real-world data; integrating insights, infrastructure, and interaction.

EDUCATION

Master of Science in Statistics - Data Science, Rutgers University, New Brunswick, NJ *Sep 2023 – May 2025*
Relevant Coursework: Probability & Inference, Data Mining, Regression, Statistical Learning, DBMS **GPA: 3.45 / 4.00**

Bachelor of Engineering in Computer Science, University of Mumbai *Aug 2019 – May 2023*
Relevant Coursework: DSA, OOPs, Software Engineering, AI, ML, NLP, Deep Learning, Big Data Analytics **CGPA: 8.5 / 10.0**

TECHNICAL SKILLS

Languages & Libraries: Python, JavaScript, R, SQL (Snowflake, SQL Server), C, C++, Java, Bash
AI/ML & NLP: Supervised/Unsupervised ML, LLM Prompting, RAG, LangChain,LangGraph, Transformers (BERT, GPT, T5)
Cloud & MLOps: AWS (S3, EC2, SageMaker, Bedrock), GCP (BigQuery, Cloud Functions)
Data Engineering: Apache Spark, Postgres, MongoDB, Apache Kafka
Visualization & Tools: Tableau, PowerBI, Streamlit, Plotly, Matplotlib, VS Code, Git, Weights & Biases

EXPERIENCE

AI Engineer and Research Assistant – *Particulate Systems Lab (PSL), Rutgers University* *Sep 2025 – Present*

- Rebuilding MATLAB-based **population balance** and **MPC** models into modular Python components to create **physics-informed ML baselines**, applying vectorization and performance tuning to achieve ∼**40% faster inference** on large-scale simulations.
- Developing automated **ML evaluation pipelines** to benchmark Python models against MATLAB ground truth, reducing validation time by **60%** and ensuring <**5% prediction error** across critical trajectory states.
- Integrating these physics models into a **digital-twin ML stack** (Streamlit + surrogate models), enabling real-time **inference**, parameter optimization, and scenario testing while profiling and improving end-to-end system latency for pharmaceutical manufacturing applications.

Software Engineer Intern – *ContinuumCare.ai* *Oct 2025 – Nov 2025*

- Developed an AI-powered, **RAG-based health assistant** using **Python**, **LangGraph**, and the **Claude API**, integrating **6+ medical guideline sources** (AHA, SSATHI, NICE, PRANA, etc.) to generate evidence-based insights with **2–3 citations per response**.
- Designed a **3-node LangGraph RAG pipeline** (profile loading → guideline retrieval → response generation) that dynamically selects guidelines based on **patient demographics** and **query intent** reducing hallucination risk.
- Built and deployed a **Streamlit web application** supporting multi-profile management, structured health data uploads (labs, vitals, history), real-time AI health summarization, and **PDF report generation**, forming the **MVP** for the company's digital health assistant.

Teaching Assistant – Part Time Lecturer & Grader – *Rutgers University* *Sep 2023 – May 2025*

- Delivered CS110/CS170 lab sessions and academic support for **200+ undergraduate students** in foundational computing.
- Taught **Scratch**, **HTML/CSS/JavaScript**, **Excel**, and **SQL queries** through live coding and interactive labs.
- Supported an **accessibility-focused** learning environment by adapting coding instruction for a **visually impaired student**, contributing to strong course performance relative to cohort benchmarks.
- Collaborated with faculty and provided feedback to improve learning outcomes, documented lab processes.

PROJECTS

Biomedical Named Entity Recognition (BioNER): (link) Built a **multi-task learning model** using **BioBERT** from **Hugging Face Transformers** to extract biomedical entities (genes, diseases, chemicals) from **5 benchmark datasets**. Engineered a preprocessing pipeline with **IOBES tagging**, token alignment, and dynamic batching to support low-resource learning. Fine-tuned shared encoders with task-specific heads, improving generalization across sparse biomedical corpora. Achieved **90%+ macro F1-score** across tasks; validated with cross-dataset evaluation and robust error analysis.

True Image Caption Generator (CNN + Transformer): (link) Built a hybrid image captioning system using **InceptionV3** for feature extraction and a dual-mode **Transformer** for sequence generation. Trained on **MS-COCO, Flickr8k, and Flickr30k** datasets with over 120K labeled images, achieving a **BLEU score of 0.80**. Designed a **Streamlit web app** for user interaction and deployed the model for real-time caption prediction from uploaded images.

CERTIFICATIONS

Certified AWS AI Practitioner *Feb 2025*
Google Cloud Program *30 Days of Google Cloud Program Certification* *Oct 2021*
Shape AI *Data Structures and Algorithms using C++ Certification* *May 2021*